

Morphology and crystal phase evolution of GeO₂ in liquid phase deposition process

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ABSTRACT

Morphology and crystal phase evolution of GeO₂ in liquid phase deposition (LPD) process is investigated. Rod-like solid phases precipitate out of solution ahead of truncated cube-like phases. SEM, XRD and TEM analyses reveal that the two sorts of solid phases are tetragonal GeO₂ and hexagonal GeO₂, respectively. The tetragonal GeO₂ phases start to experience a re-dissolving process as soon as the hexagonal phases come into being. The prior precipitation of the rod-like phase arises from a relatively low solute saturation of tetragonal GeO₂. Fast growth of a tetragonal GeO₂ phase along [111] direction leads to development of a rod-like shape. The re-dissolving phenomenon does not agree with the classic growth kinetics of crystals but is strongly favored by our calculations based on thermodynamics. The GeO₂ solutes are released in a fluctuant way by germanate ions, which promotes the occurrence of the re-dissolution phenomenon. The current researches open a door for room-temperature LPD growth of not only the hexagonal GeO₂ particles and film but also the one-dimensional tetragonal GeO₂ product.

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1. Introduction

Recently much attention has been given to GeO₂ due to its potential applications in light emitting devices [1–2], infrared waveguide [3–5], catalyst [6,7], biomedicine [7,8], vacuum technique and piezoelectric devices [9–11]. Melting, sputtering deposition, chemical vapor deposition, flame hydrolysis, electrospinning, laser ablation, physical evaporation, thermal oxidation, reversed micelles and sol–gel method have been used to prepare the GeO₂ materials [12–19].

Liquid phase deposition (LPD) is a well-known method that has been successfully used to synthesize a wide variety of oxides (TiO₂, MnO₂, Al₂O₃, ZnO, etc) and sulfides (ZnS, CdS, PbS, etc) with interesting morphology and crystal structures [20,21]. This fabrication procedure takes good advantage of a homogeneous precipitation reaction between free metal ions and anions gradually released by urea, thiocetamide, etc. The simpleness and low temperature make the LPD process much welcome both in lab and commercial scales. Efforts have also made to develop a LPD process for synthesis of a GeO₂ material. However, GeO₂ is hardly obtained using a general LPD route. Free germanium ions scarcely

exist in a liquid medium (e.g. water). It is also difficult to find an organic compound, which can slowly release anions for forming GeO₂ precipitates. Very recently, the room-temperature LPD growth of a GeO₂ product has been realized in our group by acid-induced decomposition of germanate ions in aqueous medium [4]. GeO₂ has a low solubility in water but it dissolves easily in aqueous ammonia due to formation of soluble germanate ions. Our previous work reveals that the germanate ions can be slowly decomposed into GeO₂ solutes with addition of acid. An acid-induced LPD process may occur at room temperature. This process makes it possible for the LPD growth of hexagonal GeO₂ products at low cost [4,22–24]. However, many details relevant to the growth phenomena and mechanism have not yet been sufficiently studied. For instance, our latest studies demonstrate that the LPD process does not start from the direct formation of cube-like hexagonal GeO₂ particles at all. It also involves the formation and re-dissolution of rod-like tetragonal GeO₂ phases. The detailed studies of the phase and morphology evolution of GeO₂ in the LPD process, especially in the early stage, are present in this work. The crystal growth mechanisms involved in this stage are discussed. A door is hereby opened for low-temperature LPD synthesis of not only the hexagonal GeO₂ materials but also the tetragonal GeO₂ with a one-dimensional structure. This will also be of particular interest to those who concentrate on fabrication and properties investigation of the GeO₂ materials.

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2. Experimental

Hexagonal GeO_2 powders (analytically pure), aqueous ammonia (25.6 wt%), phosphoric acid (99.8 wt%), and deionized water were used as raw materials. 100 ml deionized water was mixed with 3 ml of aqueous ammonia in a glass beaker, followed by addition of 5 g GeO_2 powders. After stirring at 70 °C for 10–15 min, a transparent germanate ion aqueous solution was obtained. The pH value of the solution was adjusted to 2 using phosphoric acid. Silica glass substrates cleaned ultrasonically in acetone and water for 30 min separately were soaked in the acidic aqueous germanate ion solution. The acid-induced LPD process was allowed to take place at room temperature for 0–24 h. GeO_2 solid phases were precipitated from the solution during this period. A substrate was taken out of the solution per half-hour. Some GeO_2 powders were collected from the substrate surface for transmission electron microscopy (TEM) analysis.

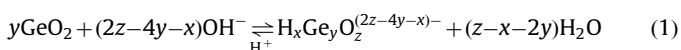
To investigate the precipitation rates of GeO_2 at different times of the LPD process, a clean glass substrate was immersed in an aqueous germanate ion solution after adding phosphoric acid for a different time. The substrate was taken out after being kept soaking in the solution for one hour. The quantity of GeO_2 particles precipitated on each substrate surface within one-hour time was recorded. The average precipitation rate over a period of

1 h was estimated according to particle quantity per unit area per hour.

The morphology of LPD-derived GeO_2 solid phases was observed using an optical microscope and a scanning electron microscope (SEM, JSM-5610LV, Japan). The detailed morphology and crystal structure were characterized by a TEM (JEOL TEM CX200). Further phase analyses were performed using X-ray powder diffraction (XRD, Rigaku D/MAX-2550PC, Cu K radiation).

3. Results and discussion

According to our previous study [4], the GeO_2 solid phases may homogeneously precipitate from an acidized aqueous germanate ion solution (see chemical Eq. (1)):



In the event that a substrate is previously soaked in the solution, the GeO_2 solid phases will be formed on the substrate surface. Fig. 1 shows SEM images of the glass substrates soaked in the solution for different times during the early stage of the LPD process (the whole LPD process can persist for more than five days). Rod-like GeO_2 solid phases with widths of 50–300 nm and lengths up to tens of

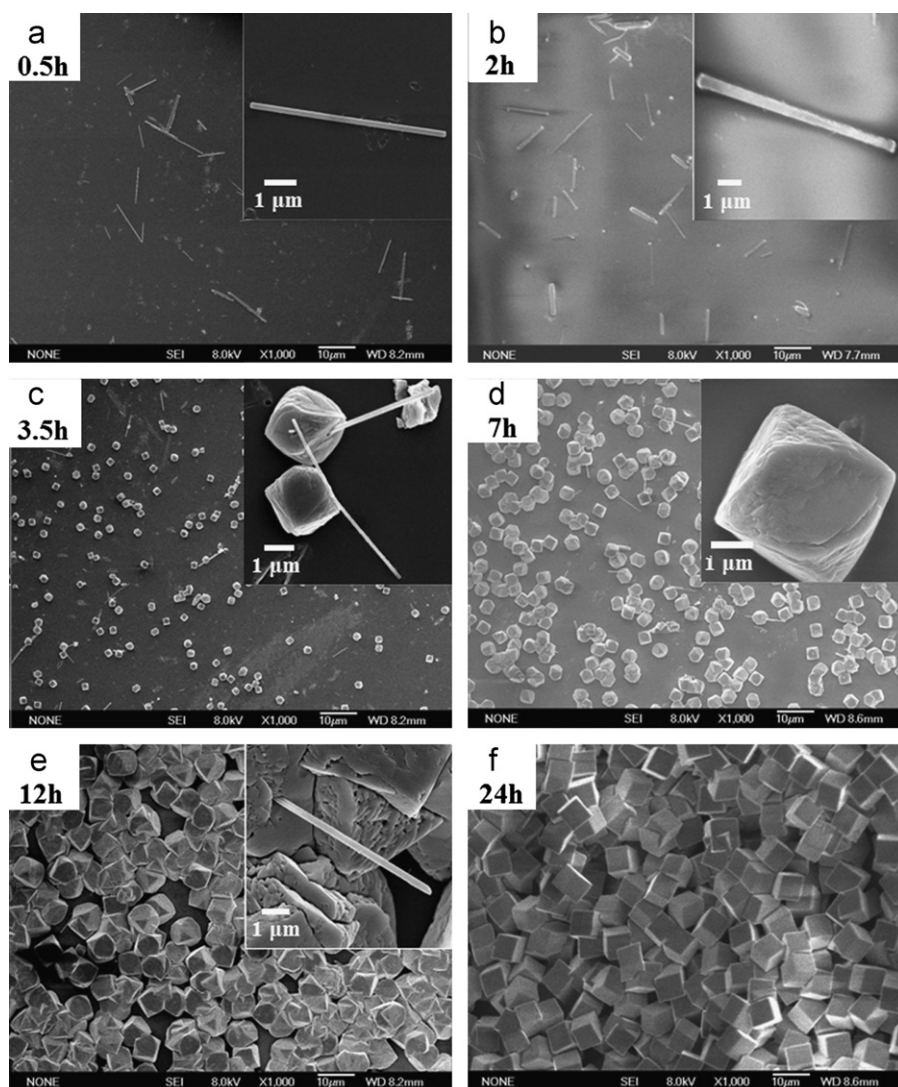


Fig. 1. SEM images of glass substrates kept soaking in the acidic aqueous germanate ion solution for 0.5–24 h.

micrometers are observed on the substrates soaked in the solution for 0.5–2 h (see Fig. 1(a) and (b)). The truncated cube-like solid phases appear on the substrate soaked for 3.5 h (see Fig. 1(c)). Although the rod-like solid phases can still be observed, their sizes have decreased to some degree. As the time goes up to 7–12 h, few rod-like phases exist while more and more truncated cube-like solid phases have kept growing up. When the LPD process proceeds for 24 h, the truncated cube-like phases have grown up into cube-like particles. The rod-like solid phases are not observable anymore.

Fig. 2 gives XRD patterns of the substrates soaked in the solution for different times. The solid phases precipitated on the substrate surfaces within 0.5–24 h are all crystallized. The observed diffraction peaks can be indexed to those of the hexagonal α -quartz GeO_2 (space group P3221) and the tetragonal rutile GeO_2 (space group P42/mnm). The solid phases on the substrate soaked for 0.5 h exhibit only a tetragonal crystal structure. The hexagonal and tetragonal GeO_2 solid phases coexist on the substrate soaked for 2–12 h. Only the hexagonal GeO_2 solid phase has been detected on the substrate soaked for 24 h. The peak intensity changes clearly with variation of the precipitation time. The diffraction intensity of the tetragonal GeO_2 phase increases at first (0.5–2 h) and then decreases (2–24 h). The peak intensity of the hexagonal GeO_2 phase keeps enhancing with increase of the precipitation time from 2 to 24 h. Based on the XRD and SEM analyses, one can see that the rod-like solid phases are tetragonal GeO_2 and the truncated cube or cube-like solid phases are hexagonal GeO_2 . The morphology and crystal structure of the solid phases were further examined by TEM analysis. Fig. 3 gives TEM images of the rod-like solid phase and the truncated cube-like solid phase. The electron diffraction patterns (see the insets of Fig. 3) disclose that the rod-like and cube-like solid phases are single crystals. Their electron diffraction patterns

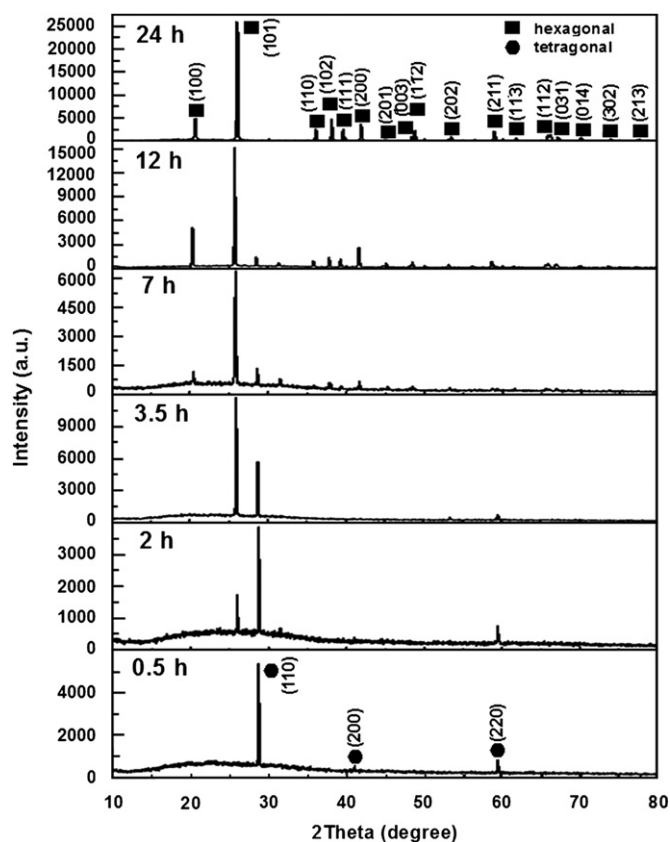


Fig. 2. XRD patterns of glass substrates kept soaking in the acidic aqueous germanate ion solution for 0.5–24 h.

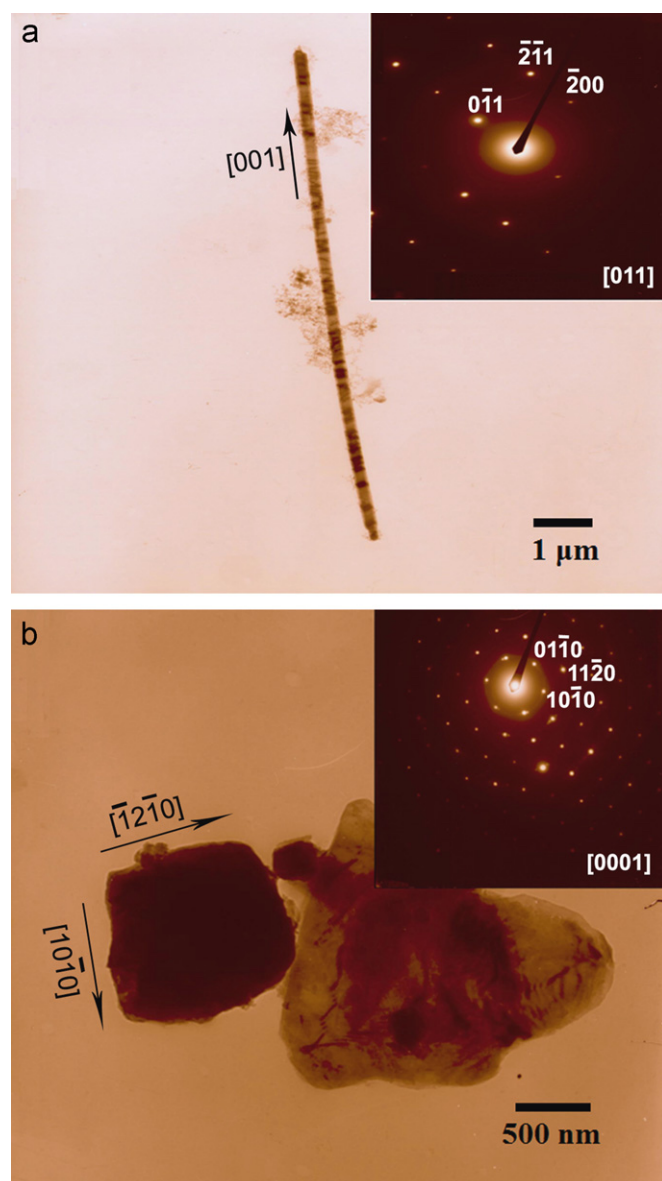


Fig. 3. TEM images of the rod-like solid phase (a) and the truncated cube-like solid phase (b). The insets are electron diffraction patterns of the two sorts of solid phases.

match with the [011] zone axis of a tetragonal GeO_2 structure and the [0001] zone axis of a hexagonal GeO_2 structure, respectively. We indexed the electron diffraction patterns and the major crystallographic directions in the TEM images are indicated (see Fig. 3(a) and (b)).

Fig. 4 describes variations between the estimated quantity of rod-like tetragonal GeO_2 phases and cube-like hexagonal GeO_2 phases along with the deposition time. The mean particle size of the hexagonal phases is also estimated. The particle size–deposition time curve is shown in Fig. 4. The particle size of the rod-like tetragonal phases includes the particle length and width. It is hard to be quantitatively estimated and presented in Fig. 4. However, the variations in particle length and width of the rod-like tetragonal phases have already been discussed on the basis of the SEM images (see Fig. 1). Fig. 4 indicates that the quantity of the tetragonal phases increases firstly (0.5–2 h) and then decreases (2–12 h). The hexagonal phases exhibit a fast increase in quantity within 2–7 h. It maintains almost unchangeable as the time changes from 7 to 12 h. The size of the hexagonal phases

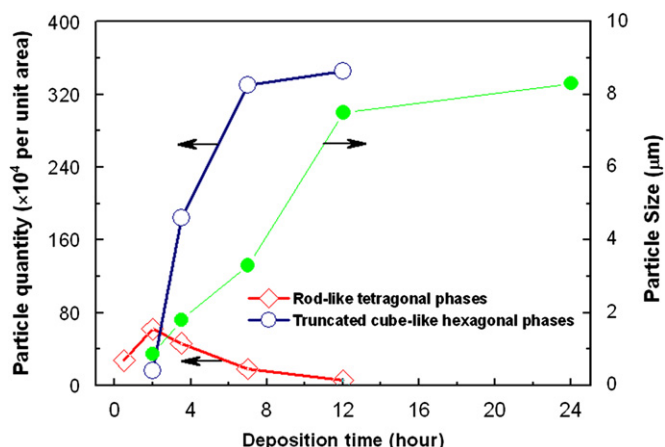


Fig. 4. Variations between the estimated particle quantities of the tetragonal and hexagonal GeO_2 phases along with the deposition time. The estimated mean particle size of hexagonal phases is also shown in this figure.

varies from 0.87 to 8.3 μm within 2–24 h. A relatively fast increase in size is observed within 2–12 h.

The above experimental results reveal several interesting facts: (1) The tetragonal GeO_2 phase has precipitated from the solution earlier than the hexagonal GeO_2 phase; (2) The tetragonal phase can develop a rod-like shape; (3) The tetragonal GeO_2 phase starts to undergo a re-dissolution process as the hexagonal GeO_2 phase appears and (4) The hexagonal GeO_2 phase keeps growing up into big ones and may finally develop a perfect cube-like shape. In the following paragraphs, possible mechanisms responsible for the facts 1 to 3 are discussed (The fact 4 has been discussed in our previous work).

According to classic nucleation and growth kinetics of crystals in a liquid medium, the nucleation rate u_1 , the growth rate u_2 , the solute concentration C and the normal solute saturation C_s have the following relationships [25]:

$$u_1 \propto (C/C_s - 1) \quad (2)$$

$$u_2 \propto (C - C_s) \quad (3)$$

Evidently, the nucleation and growth of a GeO_2 crystal occur as the concentration of GeO_2 solutes (C_{GeO_2}) is higher than its normal solute saturation ($C_{s(\text{GeO}_2)}$). There are two sorts of GeO_2 . One is tetragonal GeO_2 and the other is hexagonal GeO_2 . At room temperature (25 $^\circ\text{C}$), the tetragonal GeO_2 and hexagonal GeO_2 have normal solute saturations ($C_{s(\text{Tetrag})}$, $C_{s(\text{Hexag})}$) of 0.023 and 0.453 wt% in water, respectively [4]. With addition of acid, the germanate ions start to decompose and release more GeO_2 solutes. The C_{GeO_2} increases slowly. It will exceed the $C_{s(\text{Tetrag})}$ ahead of the $C_{s(\text{Hexag})}$ since the former is much lower than the latter. According to the relational expressions (2) and (3), nucleation and growth of the tetragonal GeO_2 crystal will take place firstly in the acid-induced LPD process. This is in accordance with the SEM and XRD results.

To understand the growth mechanism of the rod-like tetragonal GeO_2 phase, we have applied the classic Blavy law for crystal growth. The crystal shape is determined by the relative growth rates on various faces of a crystal. The general rule is that faces with the slowest growth appear as large developed faces. The Blavy law suggests that the optional growth crystal planes are those with the highest atomic packing density and the largest spacing between adjacent layers. Tetragonal GeO_2 has a rutile crystal structure (see Fig. 5(a)). The gray spheres represent Ge ions and the yellow spheres represent oxygen ions. In tetragonal GeO_2 , each Ge ion is surrounded by six oxygen ions

and each oxygen ion is surrounded by three Ge ions. The (111) crystal planes have the highest atomic packing density and the largest spacing between adjacent layers. These planes are the optional growth crystals faces compared with other planes. It is reported [26] that the crystal face with the corner of the coordination polyhedron at the interface has the biggest growth rate; the crystal face with the edge of the coordination polyhedron at the interface has the second biggest growth rate; the crystal face with the face of the coordination polyhedron at the interface has the smallest growth rate. Thus, we can judge the growth habits by making use of different elements of the coordination polyhedron at the interface. The [111] direction of the tetragonal GeO_2 crystal face with the corner of the coordination polyhedron has the fastest growth rate (see Fig. 5(b)). Moreover, if the types of elements of the coordination polyhedron at the interface are equal, the growth rate of various crystal faces is related to the number of elements. According to the rule of growth habit, one can find out that the growth rate in the [111] direction is the fastest (see Fig. 5(c)). A one-dimensional GeO_2 crystal structure is likely developed. Sharp tips are indeed observed at ends of the rod-like GeO_2 solid phases (see Fig. 5(d)). Actually, the following relationship between the velocities of crystal growth in different directions can be found: $V_{\langle 110 \rangle} < V_{\langle 100 \rangle} < V_{\langle 101 \rangle} < V_{\langle 001 \rangle} < V_{\langle 111 \rangle}$, on the basis of the rule of growth habit discussed above. The morphology evolution of the tetragonal GeO_2 phase is suggested in Fig. 5(e). The crystal plane (111) grows faster compared with other planes, which leads to formation of a tetragonal GeO_2 phase with a rod-like shape.

With progression of the decomposition of germanate ions, more GeO_2 solutes are released. The C_{GeO_2} will exceed the $C_{s(\text{Hexag})}$ sooner or later. As this happens, nucleation and growth of the hexagonal GeO_2 phase start. The hexagonal GeO_2 phase is indeed observed using SEM and XRD after the LPD process proceeds for 2 h. We have studied the morphology evolution of hexagonal GeO_2 phases in the acid-induced LPD process [11]. It is found that a hexagonal prism with two hexagonal pyramids at both ends will slowly develop a truncated cube-like shape, and then a cube-like shape [22]. Such a morphology evolution process is also confirmed by SEM images shown in Fig. 1(c)–(f). The corresponding growth mechanism need not be discussed here again. Anyway, there comes another query why the rod-like tetragonal GeO_2 phases gradually disappear as the hexagonal GeO_2 phases come forth and grow up into bigger cube-like particles.

The relational expressions (2) and (3) indicate that a C_{GeO_2} above $C_{s(\text{Hexag})}$ is necessary for nucleation and growth of the hexagonal GeO_2 phase. Meanwhile, such a C_{GeO_2} would be more favorable for nucleation and growth of the tetragonal GeO_2 phases since $C_{s(\text{Tetrag})}$ is smaller than $C_{s(\text{Hexag})}$. In this sense, the rod-like tetragonal GeO_2 phase would not disappear and might grow up into a big particle. Obviously, this is not the case observed in present work. The as-formed tetragonal GeO_2 phases have gradually re-dissolved in the solution as the hexagonal GeO_2 phases grow up. Our calculations based on thermodynamics can give a possible explanation for this phenomenon. The change in Gibbs free energy, $\Delta_r G$, between products and reactants has been widely used to judge whether a reaction process may occur spontaneously or not. We suggest that a tetragonal GeO_2 phase can transform into a hexagonal phase in an aqueous medium as indicated by Eq. (4):



The free energy change for the above phase transformation process can be calculated by Gibbs–Helmholtz Eq. (5),

$$\Delta G = \Delta_r H - T \Delta_r S = (\Delta_f H_{f,298(\text{tetra})}^\theta - \Delta_f H_{f,298(\text{hex})}^\theta) - T(S_{298(\text{hex})}^\theta - S_{298(\text{tetra})}^\theta) \quad (5)$$

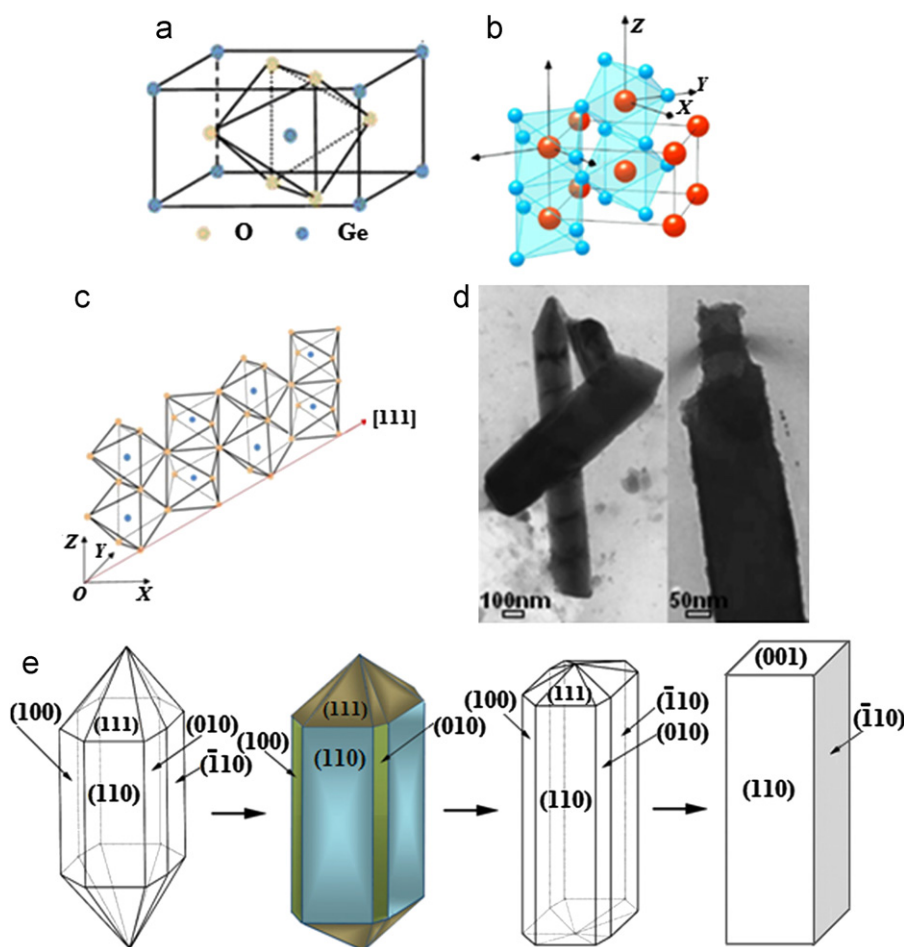


Fig. 5. Crystal structure of tetragonal GeO_2 and schematic morphology evolution of a rod-like tetragonal GeO_2 solid phase in the LPD process. (a) Crystal structure of tetragonal GeO_2 . (b) and (c) show the fastest growth rate along the $[111]$ direction. (d) TEM images showing sharp tips at the ends of rod-like GeO_2 solid phases (the rod-like samples were collected after the LPD process proceeded for 0.5–2 hours). (e) Morphology evolution of the tetragonal GeO_2 phase.

where the $\Delta_r H^\theta$ term is only weakly dependent on temperature, but the $T\Delta_r S^\theta$ term is strongly dependent on temperature due to presence of the T in the term. As for the tetragonal GeO_2 and hexagonal GeO_2 , the $\Delta_r H^\theta$ and $\Delta_r S^\theta$ values are following: $\Delta H_{f,298}^\theta(\text{tetra}) = -57,9902\text{J}$, $\Delta H_{f,298}^\theta(\text{hex}) = -76149\text{J}$ and $S_{T(\text{tetra})}^\theta = 39.748\text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$, $S_{T(\text{hex})}^\theta = 55.271\text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$ [27]. The estimated $\Delta_r G$ (about $-26176.854\text{J}\cdot\text{mol}^{-1}$) falls far below zero. That is, the further growth of a hexagonal GeO_2 phase by consumption (re-dissolution) of a tetragonal GeO_2 phase contributes to enhancing thermodynamic stability of the system. From the kinetics point of view, the re-dissolution process will be promoted if GeO_2 solutes cannot be sufficiently supplied by the mother solution. This leads us to investigating the precipitation rates of GeO_2 at different times of the LPD process. Glass substrates were immersed in the aqueous germanate ion solution after adding acid for different times. Each substrate had been kept soaking in the solution for 1 h before it was taken out. The quantity of GeO_2 particles precipitated on the substrate surface was recorded. The mean precipitation rate $\bar{u}$ (per unit area) over a period of a one-hour time was calculated. The average precipitation rate $\bar{u}$ plotted against the time is shown in Fig. 6. The precipitation rate does not increase lineally, but fluctuates with prolongation of the precipitation time. In other words, the precipitation rate fluctuates in the LPD process, at least it does so during the early stage (0–24 h). Since the precipitation rate is mainly related to C_{GeO_2} , the GeO_2 solutes must be released in a fluctuant way by germanate ions. The germanate ion has a

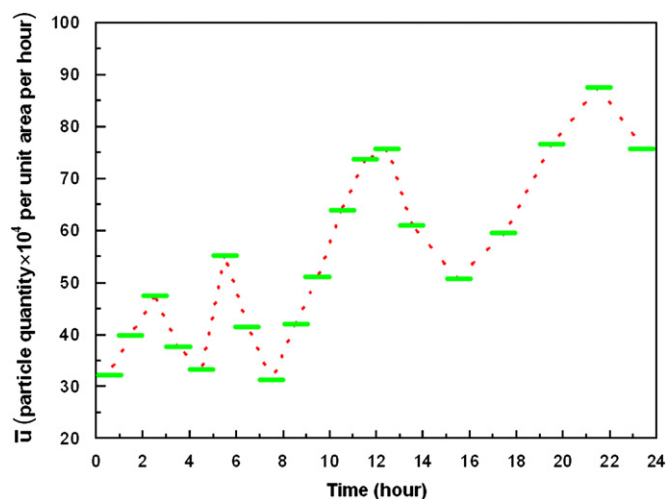


Fig. 6. Mean precipitation rate $\bar{u}$ over a period of a one-hour time at different times after adding acid to the aqueous germanate ion solution.

complex molecule structure, which makes the acidic germanate ion solution serve as a buffer solution of GeO_2 solutes. We have found that the release of GeO_2 solutes persists for as long as 5 days or so. The slow, fluctuant release of GeO_2 solutes cannot produce a steady increase in C_{GeO_2} within 0–24 h. The GeO_2 solute

may be absent to some degree within a long or short period. The LPD growth of both tetragonal GeO_2 phases and hexagonal GeO_2 phases is not allowed. The as-formed tetragonal GeO_2 phases will re-dissolve in the mother solution to provide more GeO_2 solutes needed for the growth of big hexagonal GeO_2 phases. This is favored by the fact that a hexagonal GeO_2 phase is more thermodynamically stable than a tetragonal GeO_2 phase. Anyway, the above experimental results and discussion demonstrate that the LPD growth of a one-dimensional tetragonal GeO_2 product at room temperature is possible as long as the synthetic conditions are properly controlled (e.g. a low germanate ion concentration as well as the weak acidity is needful). In this sense, the present work contributes to expanding the LPD growth of GeO_2 products from hexagonal particles or films to one-dimensional tetragonal structures.

4. Conclusions

The rod-like tetragonal GeO_2 phases precipitate out of an aqueous germanate ion solution ahead of the truncated cube-like hexagonal GeO_2 phases in the acid-induced LPD process due to its low solute saturation. Faster growth of the (111) crystal plane causes the development of a rod-like shape. The tetragonal phases start to re-dissolve in the mother solution to supply more GeO_2 solutes needed for growth of the hexagonal GeO_2 phases. The re-dissolution process is favorable for reducing Gibbs free energy of the system. This process is also furthered by a fluctuant way in the release of GeO_2 solutes. The present studies suggest that besides hexagonal GeO_2 particles and films, one-dimensional tetragonal GeO_2 products can also be grown using the acid-induced LPD process at room temperature.

Acknowledgments

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